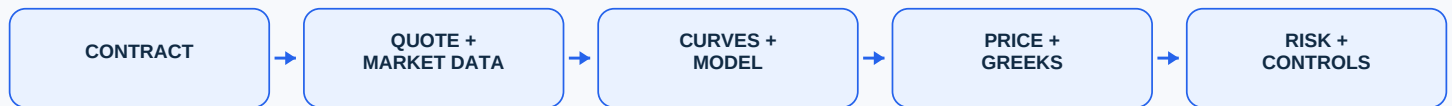


Markets, Pricing and Risk

The high-level mental model: price the contract, expose the risks, then prove the result is controlled.

QUANT INTERVIEW - 2 PAGE MAP



1. PRODUCT MAP - START WITH CASHFLOWS

Equities and forwards

Market value is $N \times S$. Holding PnL is price change plus distributions and corporate actions, less funding, borrow and fees. Under idealized continuous carry, $F_0 = S_0 \exp[(r-q)T]$.

FX

Say the quote first: base/quote means quote currency per unit of base. $F(\text{base/quote}) = S \times P(\text{base})/P(\text{quote})$. Forward points are $F-S$; NDFs settle versus a fixing without exchanging both notionals.

Fixed income and rates

$PV = \sum(CF_i \times DF_i)$. Dirty price = clean price + accrued interest. A curve discounts dated cashflows; YTM is a summary, not the curve. DV01 is the value change for a declared 1 bp rate bump. Swaps normally project floating cashflows and discount them with separate curves.

Credit

Survival under a pricing measure: $Q(0, T) = \exp[-\int_0^T \lambda(u) du]$. A par CDS balances premium and protection legs. Keep risk-neutral default curves separate from real-world PD forecasts; recovery, liquidity, basis and settlement matter.

Commodities

Baseline carry: $F_0 = S_0 \exp[(r + \text{storage} - \text{convenience})T]$. Contract month, grade, location, notice dates, limits and physical delivery can dominate the model. Signed prices are economically possible.

Inflation, financing and hybrids

Inflation products depend on exact index lags and interpolation. Repo and stock loan change executable carry. Convertibles, TRS, cross-currency swaps and structured notes combine components, but collateral, funding, legal terms and lifecycle events prevent naive additivity.

Shared quantitative foundations

- No-arbitrage links instruments through replication; pricing is a discounted expectation under a declared measure and numeraire.
- Probability supplies expectation, variance, covariance and conditional distributions; statistics turns finite data into estimates with uncertainty.
- Regression beta is benchmark- and window-dependent. Correlation is not cointegration, causality or tail dependence.
- Linear algebra drives factor models, covariance risk and optimization; calculus drives sensitivities; numerical analysis controls approximation error.
- Calibration matches selected market quotes. A good fit is necessary, not proof that dynamics or extrapolation are correct.

2. OPTIONS AND VOLATILITY

Payoff before model: call = $\max(S-K, 0)$; put = $\max(K-S, 0)$.

Black-Scholes: a benchmark under lognormal diffusion, declared rates/dividends, continuous trading and frictionless hedging - not a law of markets.

Greeks: delta = direction; gamma = delta curvature; vega = volatility sensitivity; theta = time/roll; rho = rate sensitivity. Always give the bump unit and contract multiplier.

Trading intuition: long options pay theta to own convexity; short options collect carry while warehousing gap and tail risk. Gamma hedging monetizes movement only after costs and discrete hedging error.

Surface: strike skew/smile + term structure. Enforce calendar and butterfly arbitrage controls. American exercise adds an optimal-stopping decision.

3. PORTFOLIO RISK AND PNL

Loss convention: define $L = -\text{PnL}$, currency, horizon and population before quoting a risk number.

VaR: loss quantile at confidence alpha. It says where the tail starts, not how bad it becomes.

Expected Shortfall: average of the worst 1-alpha probability mass; more tail-sensitive than VaR.

Stress: coherent named moves across spot, curves, vol, basis, liquidity and funding. Stress complements rather than replaces VaR/ES.

Extreme tails: peaks-over-threshold fits GPD excesses above a high threshold. Threshold choice trades bias against sample size; ES is finite only when shape $\xi < 1$.

CVA: discounted expected positive exposure x default loss. The product-of-marginals shortcut excludes wrong-way dependence.

PnL explain: start value + market moves + carry + trades/events + model/data changes + residual. Residual is a diagnostic.

4. NUMERICS, DATA AND CONTROL

Numerics: closed form where valid; trees/PDE for exercise and state grids; Monte Carlo for path dependence and joint exposure. Report convergence, discretization and standard error.

Market data: instrument identity, unit, quote type, source, timestamp, timezone and quality state are part of the value. Curves and surfaces must reprise their calibration instruments.

Controls: bounds, parity, monotonicity under stated assumptions, bump stability, independent benchmark, scenario replay and PnL reconciliation.

Fast sanity checks: reconcile position population and currency; test zero/expiry/extreme inputs; verify curve/surface quote units; compare local Greeks with full revaluation; investigate stale data, discontinuous calibration and unexplained PnL.

Interview line: "A model is the payoff assumptions, market inputs, numerical method and validation contract - not just the formula."

Research, Strategies and Production

Point-in-time research must survive validation, costs, execution, monitoring and replay.

QUANT INTERVIEW - 2 PAGE MAP



1. PAIRS TRADING AND COINTEGRATION

Thesis: trade a relative residual, not two standalone price forecasts. Correlation measures co-movement; cointegration asks whether a linear combination of non-stationary levels is stationary.

Canonical spread

$$s_t = \log(P_{A,t}) - \alpha - \beta \log(P_{B,t})$$

$$z_t = (s_t - \text{rolling mean}) / \text{rolling stdev}$$

Positive z : short the fitted spread. Negative z : buy it. Exit near equilibrium; use state, hysteresis, stop, time limit and break rules. The standard term is **mean reversion**, not "reverse mean."

Research-to-trade story

1. Form an economically related, point-in-time liquid universe.
2. Check integration order; fit Engle-Granger or Johansen only inside the training window.
3. Test the residual with the correct cointegration critical values; control multiple testing.
4. Estimate hedge ratio, spread volatility and a noisy half-life range; freeze parameters for the decision.
5. Convert statistical hedge into executable shares; inspect dollar, beta, sector, volatility, FX and borrow exposure.
6. Execute both legs with partial-fill controls; accrue bid/ask, impact, borrow, financing, dividends and corporate actions.
7. Monitor stationarity, hedge drift, regime changes and events; retirement is a valid model outcome.

Teaching pair: KO/PEP is intuitive but product mixes differ. Treat any example as synthetic unless live, point-in-time data proves the relationship. Cointegration is not profitability, guaranteed convergence or risk-free arbitrage.

2. FORECASTING AND MACHINE LEARNING

Time-series map

- ARIMA/SARIMA: own lags, differencing and seasonality.
- VAR/VECM: joint dynamics; VECM preserves cointegration.
- EWMA/GARCH/HAR: conditional or realized volatility.
- State space/Kalman: noisy latent state and dynamic hedge.
- HMM/change point: probabilistic regimes or structural shifts.

ML and signals

Linear/regularized models are strong baselines. Trees/boosting capture nonlinear interactions; deep sequence models need data and stability; RL is sequential control whose result is only as credible as its simulator and off-policy evaluation.

Forecast discipline

Fit only on information available at the forecast origin. Purge overlapping labels, embargo leakage paths, nest hyperparameter choice, compare with simple baselines, include costs/capacity and report uncertainty - not only one score.

Factor and systematic signals

Separate explanatory risk factors from alpha signals. Common signals include value, quality, momentum, reversal, carry, trend and seasonality. Rank cross-sectionally, neutralize intended exposures, standardize robustly and prove that turnover-adjusted returns survive universe and regime changes.

Research failure modes: look-ahead, survivorship, revised fundamentals, overlapping labels, multiple testing, overfit hyperparameters, unstable covariance, ignored borrow/costs and a backtest that cannot replay holdings from trades.

3. PORTFOLIO CONSTRUCTION

Markowitz: minimize $w' \Sigma w$ for a return target, or maximize $\mu'w - \text{risk penalty} - \text{costs}$. Means are noisy, so shrink inputs and constrain leverage/concentration.

Black-Litterman: blend an equilibrium prior with views and explicit confidence.

Risk parity: equalize covariance-based risk contributions, not weights or tail losses.

Kelly: maximize expected log wealth; estimation error and drawdown motivate fractional Kelly.

HRP: cluster dependence and allocate recursively; avoids a full inverse but still depends on window/linkage.

Always add: turnover, borrow, liquidity, limits, factor exposure, scenario loss and rebalance timing.

4. EXECUTION, FUNDING AND TCA

Execution objective: convert target positions into fills while balancing urgency, spread, impact, volatility and opportunity cost.

TWAP is time-weighted; VWAP follows a volume profile; POV caps participation. Implementation shortfall compares decision price with executed economics. TCA separates spread, timing, impact, fees and residual. Borrow and funding can erase a paper edge.

5. PRODUCTION QUANT ENGINEERING

Point-in-time data: as-of timestamps, effective timestamps, lineage, revisions and immutable event replay prevent look-ahead bias.

Architecture: typed trade/model objects; versioned market snapshots; separate scenario generation, valuation and aggregation; deterministic replay.

Testing: unit identities, edge cases, benchmark prices, property tests, regression tests, performance tests and production reconciliation.

Lifecycle: economic agreement, confirmation status and cashflow settlement are separate states. Handle fixings, exercise, margin, corporate actions, novation, termination and failures.

Governance: model inventory, approval, limitations, reserves, monitoring, change control, observability and rollback.

6. HOW TO TELL THE STORY IN AN INTERVIEW

Use this six-part answer:

1. Objective and economic intuition.
2. Contract, sign, units and timing.
3. Data, assumptions and model equation.
4. Hedge, risk or portfolio decision.
5. Failure modes and alternatives.
6. Validation, execution and monitoring.

Close with: "I would test the simple benchmark first, then add complexity only if it survives out of sample and after costs."

High-value contrasts

- Price vs PnL: snapshot value vs change plus cashflows.
- Risk-neutral vs real-world: valuation weights vs forecasts.
- Correlation vs cointegration: co-movement vs stationary relation.
- VaR vs stress: probabilistic quantile vs coherent named scenario.
- Target vs execution: desired weights vs fills, costs and funding.
- Fit vs validity: matching history is not out-of-sample evidence.